

Pandas Practice Assignment: Nobel Prize Dataset

Objective

Practice using Pandas by exploring and analyzing the Nobel Prize dataset. Students should use DataFrame operations, filtering, grouping, aggregation, sorting, and basic data analysis techniques to answer the following questions.

Dataset

Use the Nobel Prize dataset provided by the instructor. You are encouraged to explore the dataset structure before starting the analysis.

Requirements

1. What is the most commonly awarded gender and birth country?
 - Store your answers as string variables named `top_gender` and `top_country`.
2. Which decade had the highest ratio of US-born Nobel Prize winners to total winners in all categories?
 - Store your answer as an integer variable named `max_decade_usa`.
3. Which decade and Nobel Prize category combination had the highest proportion of female laureates?
 - Store your answer as a dictionary named `max_female_dict` where the decade is the key and the category is the value. The dictionary must contain only one key:value pair.
4. Who was the first woman to receive a Nobel Prize, and in what category?
 - Save your answers as string variables named `first_woman_name` and `first_woman_category`.
5. Which individuals or organizations have won more than one Nobel Prize throughout the years?
 - Store the full names in a list named `repeat_list`.

Guidelines

- Use only Pandas for the analysis (no manual calculations).
- Explore the dataset using functions such as `head()`, `info()`, `describe()`, and `value_counts()`.
- Write clean, well-commented code.
- Use meaningful variable names and follow the required variable names exactly.
- Try to solve each task using efficient DataFrame operations such as `groupby()`, `sort_values()`, filtering, and aggregation.

Deliverables

- A Jupyter Notebook (.ipynb) containing all analysis steps and answers.
- All required variables must be created with the exact names specified in the assignment.
- Include brief explanations or comments describing your approach.